The health consequences of workers in healthcare systems managing Cov-Sars-19 outbreaks in Spain.

The impact of Covid-19 on mental health amongst healthcare workers while working during a pandemic.

Michella Voughn Warren

MASTERS IN RESEARCH IN SOCIOLOGY AND DEMOGRAPHY

UNIVERSITAT POMPEU FABRA

Universitat Pompeu Fabra Barcelona

MIND COVID

IMIM Parc de Salut MAR Barcelona

Institut Hospital del Mar d'Investigacions Mèdiques

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ABSTRACT

I did a quantitative analysis and simple exploration of possible mental health disorders that individuals are developing while living & working through the Covid-19. The methodology comes from the basis of another study conducted from the MINDCOVID research group in Barcelona, Spain, which sought out to see the mental health effects on health care professionals working in facilities where they were dealing with infected patients. With primary data used from their initial survey sent out on May 2020, I examine variables from the MINDCOVID[1] survey, in order to do a simple exploration in terms of the possible effects of increase or decrease of anxiety, PTSD, panic attacks, pre-and post COVID infection. As their study analyses general status in the realm of mental health with health care workers, there are three other groups that are being analysed along with the sample that I have received. From the data presented, around 34% have experienced anxiety at higher levels than normal. Also, an alarming percentage of over 22% of them showed signs of early PTSD, and another 22% having had had at least one panic attack in the last 30 days. Moreover, over 75% of the participants of health care workers gave no response as to whether they felt they could benefit from any additional psychological support of any kind. Lastly, there was a small percentage of around 8% of participants who admitted to having suicidal thoughts and even a few others who had tried to take their own life within the last 30 days from the initial survey date in May. My conclusion is the call for further studies within the general public and how we can solve this other potential global dilemma in mental health issues that will arise from Covid-19 apart from actually contracting the virus.

[1] https://www.mindcovid.org/inicio

I - INTRODUCTION

According to Anthony Fauci, MD. and director of the National Institute of Allergy and Infectious Diseases in United States, the COVID-19 pandemic is the worst to hit us in over 100 years. As the numbers are still rising, there are already over 41 million cases and over 1.1 million deaths worldwide. While its seems that Spaniards are reliving what seems to be a horrible repeat of the Spanish Flu, but this time supposedly more equipped, we are seeing that mental health is still underrated. Mental health diagnoses during the year 2020 are sure to have major fluctuations as the situation continues. While countries are racing to get an effective vaccine, other issues are in desperate need of attention meanwhile. Let's remember the intense mental stress that is occurring due to the magnitude of such a situation. Psychologists and other specialised physicians have recently come out with studies saying that individuals who've come into contact with the virus, have indeed, developed depression, anxiety, and higher levels of stress (Wang C., Pan R., Wan X., 2020). There is no doubt that healthcare workers are suffering, but the question is to which extent? As we are still trying to understand all of the pro-longed secondary effects of those who've been infected, we also need to take into account the gravity of those who are suffering that are not infected. The situation of a pandemic can be classified as a sort of trauma, especially in how we perceive and process it in the brain. The importance of this will be further discussed later in the background. The main issue I present is how exactly is this pandemic going to affect individuals globally from a mental health perspective? Does one really know the impact that this is having on workers in the health care facilities who are treating infected patients in quarantine? A pandemic requires another procedure in order to avoid possible spread of such a contagious virus, and that is indeed a burden nobody would want to be responsible for, if it did indeed is spread due to negligence or ineffective protocol. While it's still being studied how the diffusion of the virus is occurring, the purpose for this study is to show the importance for further research and a call for professionals to address the growing issue of these mental health diagnoses bilaterally before it, too, also gets out of hand.

Is the Covid-19 pandemic a trigger for post-traumatic stress disorder, anxiety and stress for everyone currently in Spain?

Apart from the impact on mortality and immediate physical symptoms stemming from Sars-Cov-19 virus, there is still much to be examined on the overall mental health effects on individuals globally. A study was recently conducted in Spain, which mentioned the findings that over half of healthcare professionals from Spain were at a high risk in chances of suffering from a mental disorder after the first initial breakout in March 2020 (Alonso, Mortier, Vilagut, et al.; project MINDCOVID; 2020). They also mention the increased risk and prevalence within the sample observed of suicidal thoughts within the same sample. While this analysis isn't about addressing already known causes or about what mental health is, I feel it's important to give a bit of context in order to understand how the recent and prolonged situation in Spain during the recent

Sars-Cov-19 outbreak, is a trigger for those who haven't been diagnosed. Also, it is to see any possible correlations and potential heightening levels of episodes for those had already been diagnosed with mental disorders pre-covid initial outbreak dates in March of 2020.

What's mental health got to do with it?

Truth be told, it may not be conclusive enough with the data sample from the survey that I have been given, however, we can point out some interesting findings in terms for how Covid-19 is affecting the particular situation for healthcare workers in Spain. I am thankful to have come across a project called MINDCOVID, which arose from a team of researchers and professionals at the IMIM (Institut del Mar d'Investigacions Mèdiques) Institute in Barcelona, Spain. Their team focused on creating a controlled cohort study which consisted of an anonymous survey with four different groups for intervals of four months, in order to see the effects of mental health changes of each group during the pandemic. The team collaborated with over twenty research groups throughout 6 autonomous communities in Spain. The other collaborating teams are associated with CIBER (Consorcio de Investigación Biomédica en Red) of Epidemiology and Public Health, Mental Health and other networks; they included professional experts in Public Health, Psychiatry and Mental Health, and in Occupational Medicine and primary Care. The MINDCOVID project is financed by the ISCII[2], which stands for the Instituto de Salud Carlos III; Ministerio de Ciencia E Innovación. This is my source for the primary data that I acquired in order to continue out the study.

II - BACKGROUND

A. Theoretical framework

Mental health is discussed globally as it as a global issue amongst many individuals around the world. While the way mental health is measured is debated amongst doctors and researchers in order to determine treatment for those in need, there are many scales that have proven to give us a feasible quantitative measurement in order to see possible causal relationships. While there are many figures and studies already on the amount of mental health disorders out there, we focus on depression, anxiety and post-traumatic stress disorder(PTSD) as they are currently the top three most common mental disorders globally to date. The World Health Organisation published in a recent report that over 264 million people suffer from depression alone (2020). It is no mystery that depression can also lead to the number of suicides, as it is a contributing factor to potential suicides. Anxiety disorders are also amongst the most common mental disorders. Around 284 million people were diagnosed with anxiety disorders in 2017(IHME[3],2017). I define PTSD by the definition given from the American Psychiatric Association, which defines PTSD as a psychiatric disorder that may occur in people who have experienced or witnessed a traumatic event

[2] https://www.isciii.es/Paginas/Inicio.aspx
[3] Institute for Health Metrics and Evaluation

such as a natural disaster, a serious accident, a terrorist act, war/combat, or rape or who have been threatened with death, sexual violence or serious injury (2013).

What is important to grasp from all of this, is the important question of who is exactly at risk for mental disorders? Essentially anybody can develop one throughout one point of their life, depending on the sequence of events and life one lives. There are of course mild cases which are temporary to the other extreme of more chronic diagnoses depending on the patient's case. So, with the current pandemic Covid-19 around us, this brings another important question; how does the mind digest and process this? I make this event as a dependent variable for this study and make the assumption that it is a traumatic event for all, which is important to know when it is mentioned in the equation in terms for its relevance to how it is potentially affecting our mental state. Also, to what degree it is affecting individuals is very important. I want to quickly list the possible factors which can contribute to an increase in triggers of mental disorders;

- Childhood abuse, trauma, or neglect
- Social isolation or loneliness
- Severe or long-term stress
- Unemployment or losing your job
- Death of a loved one
- Social disadvantage, including poverty or debt
- Violence of any sort
- Drug and alcohol misuse
- Significant trauma, such as military combat, war, or being involved in a serious accident and so forth.
- Lifestyle factors (diet, sleep patterns, and so forth)

Is Covid-19 triggering PTSD symptoms? First, I shall give the definition which is defined as one having been exposed to or to have witnessed an event that is threatening to safety; and one must respond to this event with fear, horror, or helplessness. Secondly, one must report a re-experiencing symptom, which may include intrusive memories, nightmares, a sense of reliving the trauma, or psychological or physiological distress when reminded of the trauma (Wang C., Pan R., Wan X., 2020). Third, there need to be at least three avoidance symptoms, which can include active avoidance of thoughts, feelings, or reminders of the trauma, inability to recall some aspect of the trauma, withdrawal from others, or emotional numbing. Lastly, one must suffer marked arousal, which can include insomnia, irritability, difficulty concentrating, hypervigilance, or heightened response. These symptoms must cause marked impairment to one's functioning, and then, can technically only be diagnosed when they are present at least 1 month after the supposed trauma in order to be classified as suffering from PTSD.

B. Hypotheses

- **H1**: Individuals who are working in health care facilities that are managing the pandemic are impacted with higher levels of stress and anxiety.

- **H2**: Those individuals who have been diagnosed with previous mental health conditions, have shown an increase in episodes whilst working through the pandemic in their health care facility.

- **H3:** Health care professionals (HCP) have developed possible mental health challenges since the initial start of the COV-SARS-19 pandemic (given from the data results).

- **H4**: With time, the symptoms are more trigged by the initial stressor, COVID, and the symptoms are intensified. (PTSD, anxiety, episodes of depression, thoughts of suicide, restlessness at night/insomnia).

- **H5:** Many respondents who've participated in the survey left many questions blank with no response, perhaps due to embarrassment or shame socially of potentially be labelled with mental disorders.

- **H6:** Those HCP's who have had previous mental health conditions prior to the initial start of the covid pandemic, have had more episodes or had an increase in their condition as they have intensified; this would be from the propaganda/worrying of getting infected and/or infecting others.

III - LITERATURE REVIEW

MINDCOVID

The MINDCOVID study that was conducted had initially collected this data through outsourcing through a contracted telephone company, which then would call those who were registered within the hospitals asking them if they would like to participate in the study which were to collect data for the general public (Alonso, et al., 2020). The research group collected data for the main groups, general public, HCP, and infected patients. For this specific analysis, I am using the data sample for health care professionals, which was also pulled from 19 different hospitals throughout all of Spain. I have the primary data from the first initial survey sent out in May of 2020. They initially gathered contestants through epidemiologic surveillance services,

hospitals, and primary care centres in 6 Spanish Autonomous Communities. Another study done that I used as my basis for using covid-19 as a stress level effector, comes from a list of the fears that were revealed that are related to the situation as a whole and contribute according to the researchers as stressors. The danger and contamination, fear of adverse socio-economic consequences, checking and reassurance seeking, xenophobia (discrimination against foreigners) and traumatic stress symptoms (for example, pandemic-related nightmares), are stress heightening factors in their COVID stress scale (Gordon, 2020) and reveal that it is a feasible tool in terms of measurement of added stress triggers.

In a Cambridge study done recently on COVID-19 (Galbraith, Boyda, Hassan et al., 2020), they mention the effects from previous outbreaks such as SERS & MARS back in 2012, and the effects that health care professionals were having. They argue that doctors experience high levels of work stress even under normal circumstances, but many would be reluctant to disclose mental health difficulties or seek help for them, with stigma being an often-mentioned reason. The apparent corona virus emergency places additional pressure on doctors and on the healthcare system in general and their research shows that such pressure brings a greater risk of psychological distress for doctors and HCP's as well. The broad range according to the doctors were a broad range of psychological morbidities, including trauma, which could potentially endure for many months after the outbreak they claimed.

IV – METHODOLOGY

This analysis utilizes survey data collected from a research group, known as MINDCOVID, involved at the IMIM centre and Hospital del Mar in Barcelona, Spain (Alonso, Mortier, Ferrer et al., 2020). The sample given consisted of workers in healthcare systems from 19 different hospitals throughout all of Spain. I used an approved list of variables given to me which was then analysed in order to answer the research question and hypotheses along with a codebook (please refer to appendix for list; figure 1.1). The data was collected during 2020 and is still in effect. This sample consisted of 9,138 participants who had answered the questionnaire via an online survey that was sent out through email. As the original researchers developed their study to analyse general status in the realm of health care workers working in facilities facing patients with Sars-Cov-19, they also have sample data of other cohort groups to see possible effects on those other than HCP's. The MINDCOVID study model was based on a cohort observational model. Mentioning this is essential in order to understand the premises of how the survey questions were formed and structured. While the research group have drawn their own conclusions, my own conclusions are being drawn by analysing primary data from the variable list. I manipulated the data in excel and STATA and have drawn conclusions from regression modelling and multivariate analysis as well as individually manipulating each variable with the complete list of participants.

My dependent variable is Covid-19 itself, not as the actual virus, but Covid-19 as a social factor which is causing a shift globally in society. My independent variables that were examined were;

- Gender (A2)
- Marital status (A8)
- Those who took a PCR test (nasal hyssop or laboratory test) (B1)
- Confirmation of Cov-Sars-19 (B2)
- Problems with anxiety (nervousness, worries, fears, obsessions (C1_4)
- Frequency of feeling nervous, tense or distressed in the past 14 days (C3_1)
- Frequency of constant worrying and not being able to stop in the past 14 days (C3_2)
- The amount of panic attacks in the last 30 days (C5)
- In the last 30 days, have you ever wished you were dead or able to sleep and never wake up again? (C_10)
- In the last 30 days have you ever thought of how you would commit suicide (e.g. taking pills or throwing oneself out a window) or have you planned to do it? (C_13)
- Approximately how many times in the past 30 days (desires or thoughts)? (C_14)
- In the last 30 days, have you committed a suicide attempt (i.e. have you deliberately injured yourself with at least any intention of dying)? (C_15)
- Receiving of psychological support in a lifetime (D1)
- Receiving of psychological support or medication in the past 12 months (D2)

For the relationship between traumatic life events, we use the dependent variable of covid-19 and compare it to our independent variables such as suicide, PTSD, anxiety and trauma from this sort of disaster events; which can increase suicidal ideation in emergency workers. Each variable had a simple (0) or (1) response, 0 being no and 2 being yes. This is referred to as hot encoding in order to quantify qualitative data from survey questions. There were a few other independent variables, such as C3_1 and C3_2, which used a scale from 1 to 5; 1 being not at all, to 5, being every day. Below is the complete list of variables ran through in STATA.

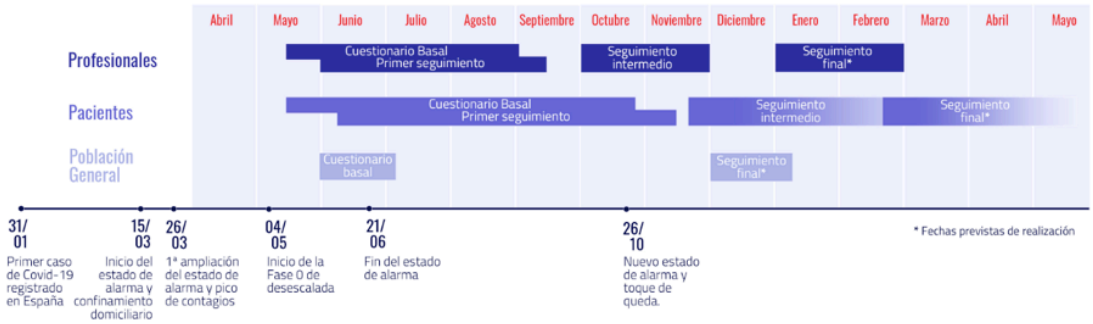
Variable	Obs	Mean	Std. Dev.	Min	Max
a1	9,138	45.03272	11.511	18	75
a2	9,126	1.806816	.3948179	1	2
a7	9,118	.7064049	.4554336	0	1
a8	9,123	1.738025	.6777285	1	4
b1	9,136	.8247592	.3801937	0	1
b2	7,516	.1930548	.3947219	0	1
b3	7,681	.0303346	.1715174	0	1
b5	1,681	.0660321	.2484121	0	1
b16_4	9,114	2.955124	1.239314	1	5
c1_1	9,019	.1088813	.3115074	0	1
c1_2	9,019	.0038807	.0621777	0	1
c1_3	9,019	.0357024	.1855574	0	1
c1_4	9,019	.348043	.4763762	0	1
c1_5	9,019	.0090919	.0949224	0	1
c1_6	9,019	.0064309	.0799389	0	1
c1_7	9,019	.0284954	.1663925	0	1
c1_8	9,019	.5873157	.4923442	0	1
c3_1	9,129	2.020922	.8718881	1	4
c3_2	9,127	1.856689	.9000298	1	4
c3_3	9,123	2.011729	.8833365	1	4
c3_4	9,128	2.072962	.918827	1	4
c3_5	9,124	1.514796	.7691437	1	4
c3_6	9,126	2.011944	.8729542	1	4
c3_7	9,128	1.682296	.843946	1	4
c4_1	9,125	1.888329	1.047387	1	5
c4_2	9,119	1.810835	1.047286	1	5
c4_3	9,125	2.113973	1.152046	1	5
c4_4	9,122	1.862201	1.037917	1	5
c5	8,885	.9465391	3.130682	0	50
c10	9,099	.076382	.265623	0	1
c11	9,102	.026258	.1599103	0	1
c13	735	.2693878	.4439438	0	1
c14	191	4.183246	5.12814	0	30
c15	9,094	.0008797	.0296483	0	1
d1_1	9,016	.2243789	.4171958	0	1

I had the complete list of these variables, in an excel sheet, and ran them through STATA first to get a general summary of each one. For the categories of age, I had to go through this separately as the age difference varied from 18 to 75 years of age. (Please refer to appendix 1.2 for questions in their original form in the survey). I ran a multilevel model in the end using variables for treatment use to see any significance (tx_lt, stb_30dd, tx_12m). As you see below, this only accounted for nearly 25% percent of the total number of observations.

```
. regress b16_4 tx_lt stb_30dd tx_12m
note: tx_lt omitted because of collinearity

      Source |       SS           df       MS      Number of obs   =     2,266
-------------+----------------------------------   F(2, 2263)      =      5.27
       Model |  16.452162         2  8.22608098   Prob > F        =    0.0052
    Residual |  3533.53195     2,263  1.56143701   R-squared       =    0.0046
-------------+----------------------------------   Adj R-squared   =    0.0038
       Total |  3549.98411     2,265   1.5673219   Root MSE        =    1.2496

------------------------------------------------------------------------------
       b16_4 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       tx_lt |          0  (omitted)
     stb_30dd |   .1148681   .0357485     3.21   0.001     .0447648    .1849714
       tx_12m |   .0052359   .0536753     0.10   0.922    -.1000221    .1104939
        _cons |   2.964325   .0349608    84.79   0.000     2.895766    3.032884
------------------------------------------------------------------------------
```



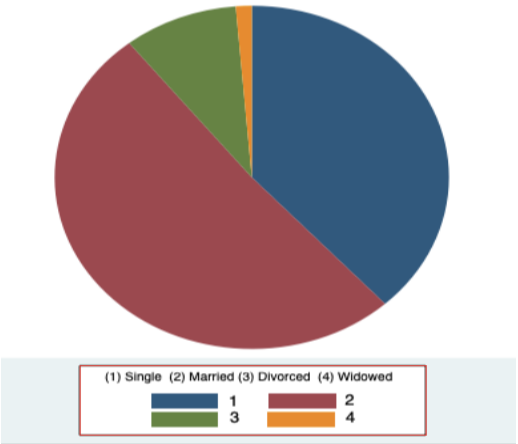
The timeline above is from the initial study, as it shows the methodology for their dates of sending out the survey to all three cohort groups (MINDCOVID, 2020).

V - FINDINGS/ RESULTS

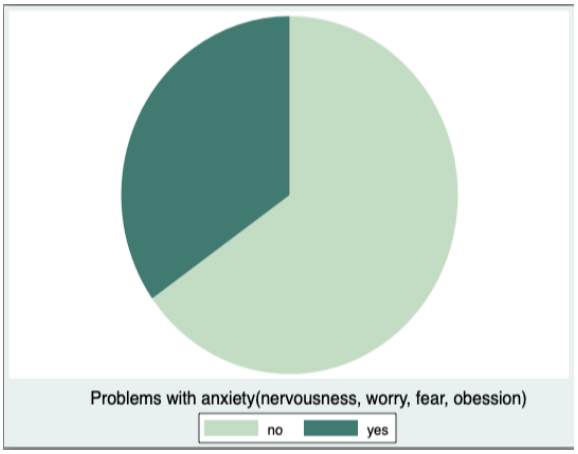
Half of the observations are married, which you can see in pie chart 1 below. From the data presented, around 34.4% have experienced anxiety at higher levels than normal (refer to pie chart 2), using variable C1_4. Furthermore, another 22% had at least one panic attack in the last 30 days, shown with variable C5. Also, an alarming percentage of over 22% of them showed signs of early PTSD (shown in regression above with variables tx_12m, tx_lt). Over 75% of the participants of health care workers gave no response as to whether they felt they could benefit from any additional psychological support of any kind (D2_2). Pie chart 3 represents the amount of observations who responded to having suicidal thoughts in the past 30 days while chart 4 demonstrates how many days in the past 30 days have you had these suicidal thoughts. Lastly, there was a small percentage

of around 8% of participants who admitted to having suicidal thoughts and even a few others who had tried to take their own life within the last 30 days from the initial survey date in May (variable C_15).

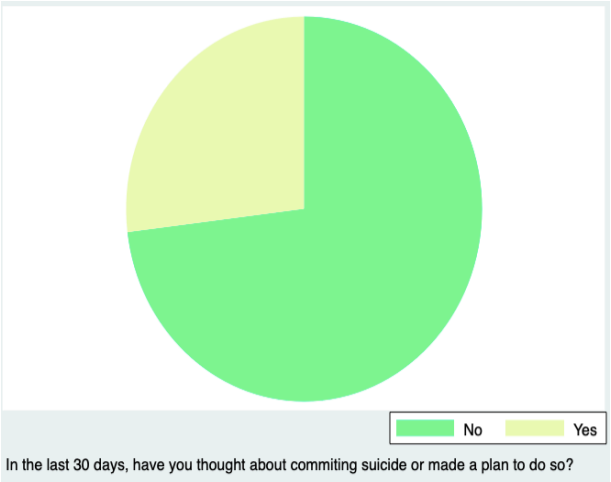
1.



2.



3.



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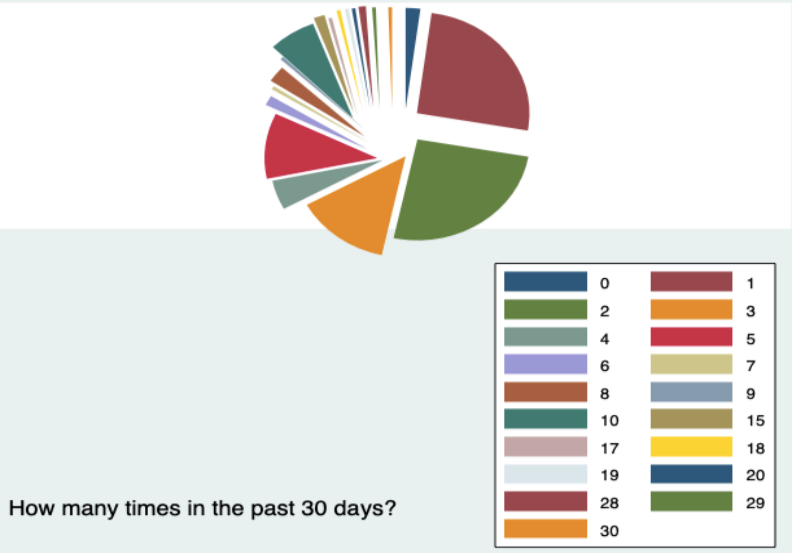
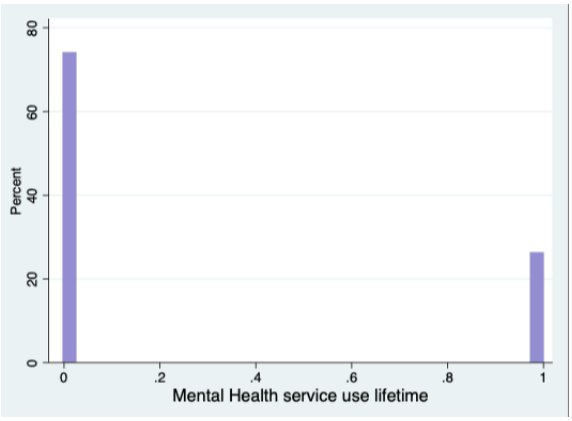
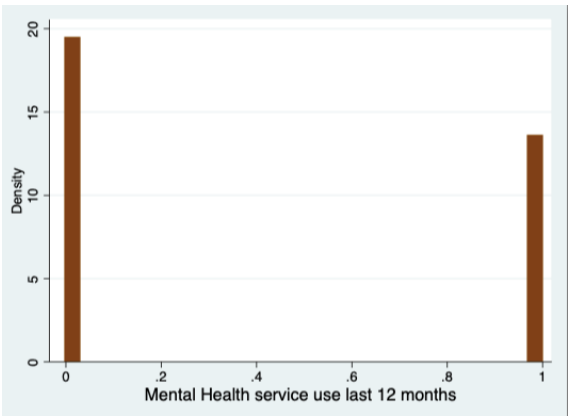


The overall findings with the variables showed that the average age for the HCP's were around the age of 38 years old. Many of the observations mentioned not having had been diagnosed previously with a mental disorder prior to the arrival of Covid-19. Interestingly enough, when you cross check variables C3_2(unable to stop worrying or not being able to control being worried) with C1 variables (symptoms pre-Covid arrival), you see that the observations who'd been pre-diagnosed had less signals of nervousness than others in the last 30 days. All observations showed signs of pre-PTSD triggers when following variables GAD_2w, panic_30dd and ptsd_30dd (appendix 1.2). Bar charts 5 and 6 (below) show the amount of observations that had utilized mental health services pre-Covid arrival (variable TX_LT), versus mental health solicitation post arrival in March 2020(variable TX_12m). The results for these two variables showed that nearly 25% of observations had used services prior to the pandemic while 10% admitted to soliciting services afterward. An important thing that should be mentioned is that for the variable, TX_12m, there was only a grand total of 2285 responses. This discrepancy shouldn't be overlooked, as the potential sociocultural prejudices could potentially play a part with those not wishing to share this information and so forth. When we look at variable B3(Any medical professionals diagnosed you with Cov-Sars-19), it shows that only 233 of those 7535 observations (variable B1) who'd taken a PCR were actually diagnosed with having the virus. This for some is another discrepancy that shouldn't be overlooked, as potentially it could be due to lack of protocol and follow-up with HCP's. More research needs to be done to see any possible causal reasons for this and the potential consequences for not having a proper diagnostic while working in these facilities, as uncertainty can linger in ones' day to day.

5.



6.



VI - CONCLUSION

While the intention was to conduct a simple analysis of primary data from a survey conducted on mental health during the pandemic in Spain, it seems that it is much more complex than a simple analysis with observation sample analysis. When we look back out initial hypotheses, one could conclude that H1, H2, H3, H4 and H5 have substantial support from the data provided. H6 which referred to HCP's who'd been previously diagnosed prior to the pandemic, still needs further follow up from those specific observations to really find it causal. The reason for this is as they've been previously diagnosed, there needs to be a more defined difference in measurement of which level is their level pre-arrival versus now. Also, for those who'd been infected with Cov-Sars-19 and were diagnosed previously, one would need to further investigate these cases to see the fluctuations in their behavior to see if its solely from the presence of the virus, and also the possibility of a symptom post infection. We can see individuals from the observations who are working in health care facilities that are managing the pandemic are indeed impacted with higher levels of stress and anxiety. We can conclude to some degree that those individuals who have been diagnosed with previous mental health conditions, have shown an increase in episodes whilst working through the pandemic in their health care facility. With time, we see that there is significance in symptoms that are provoked and trigged by the initial stressor, COVID, and the symptoms are intensified (PTSD, anxiety, episodes of depression, thoughts of suicide, restlessness at night/insomnia). My last concern is the amount of respondents who've participated in the survey had left many questions blank with no response, perhaps due to embarrassment or shame socially of potentially being labelled with mental disorders. However, this requires more research in order to make this sort of assumption as I am basing it off of the no response percentage from each question.

In the coming years, governments will not only have to deal with economic problems and unemployment, but also perhaps with a bigger problem, the mental health of their population and healthcare workers. Covid-19 is taking a toll on mental health as there are studies being conducted currently with some findings that it has already impacted society. As mentioned above, depression and anxiety and increases in cases of PTSD have increased according to recent studies conducted by other facilities. While there are other possible factors that can increase this may be due to limitation of mobility, and not being able to have social gatherings, this is only one part. National legislation plays a very important role in being able to implement opportunities and solutions for citizens in Spain. Implementation levels at the national level are still up for discussion in Spain, as there has not been anything in this regard for the public yet. While there are other services from NGO's that have a free call line for psychosocial support, such as Red Cross, this is a service proposed from RC, not the government directly. My final conclusion is the call for further studies within the general public and how we can solve this other potential global dilemma in mental health issues that will arise from Covid-19 apart from actually contracting the virus.

VII - APPENDIX

1.1

Los variables que he pedido con los siguientes:
SECTION A. SOCIODEMOGRAPHIC VARIABLES
• AGE
• GENDER
• LIVING WITH A PARTNER
• MARITAL STATUS
SECTION B:COVID-19 INFECTION STATUS:
• POSITIVE COVID-19 TEST AND/OR MEDICAL DIAGNOSIS OF COVID-19
• DEGREE OF WORRY OF FAMILY AND FRIENDS OVER GETTING INFECTED THROUGH THE HCW
SECTION C. MENTAL HEALTH
PRE-PANDEMIC LIFETIME MENTAL DISORDERS
• POSITIVE SCREEN FOR GENERALIZED ANXIETY DISORDER
• POSITIVE SCREEN FOR POST-TRAUMATIC STRESS DISORDER
• 30-DAY PANIC ATTACK(S)
• SUICIDAL THOUGHTS AND BEHAVIORS
SECTION D. TREATMENT USE
• PAST YEAR TREATMENT USE (PSYCHOLOGICAL AND/OR PHARMACOLOGICAL TREATMENT)
• PERCEIVED NEED FOR (ADDITIONAL) PSYCHOLOGICAL AND/OR PHARMACOLOGICAL TREATMENT DURING THE COVID-
19 PANDEMIC (D5/D6)

1.2 Survey questions

name	variable label	categories	Value labels / range	Skip logic (subsample the item is asked)
A1	¿Cuál es su edad?	.	(0-110)	
A2	¿Con qué genero se identifica?	1	Masculino	
		2	Femenino	
		3	Otro	
A7	¿Está viviendo en pareja?	0	No	
		1	Yes	
A8	¿Cuál es su estado civil legal?	1	Soltero/a	
		2	Casado/a	
		3	Divorciado/a o separado/a lega	
		4	Viudo/a	
B1	¿Se ha hecho alguna vez la prueba (por ejemplo, test de laboratorio o hisopo nasal) para coronavirus?	0	No	
		1	Yes	
B2	¿La prueba ha confirmado que tenía coronavirus? Si le han hecho la prueba en varias ocasiones, ¿alguna de esas pruebas ha confirmado que tenía coronavirus?	0	No	Esta variable tiene un salto: sólo se responde si B1 = YES
		1	Yes	
B3	¿Algún profesional médico le ha diagnosticado coronavirus?	0	No	Esta variable tiene un salto: sólo se responde si (B1 = No or Missing) or (B1 = YES and B2 = NO or missing)
		1	Yes	
B5	Ha estado hospitalizado/a por coronavirus?	0	No	Esta variable tiene un salto: sólo se responde si B2 = YES or B3 = YES
		1	Yes	
B16_4	Desde la llegada de la pandemia del coronavirus a España, Mi familia y amigos estaban preocupados de contagiarse a través de mí.	1	Nunca	
		2	Sólo una vez	
		3	Algunas veces	
		4	Casi siempre	
		5	Siempre	
C1	**Antes de la llegada de la pandemia del coronavirus a España, ¿alguna vez tuvo alguno de los siguientes problemas psicológicos?**			
C1_1	Depresión	0	No	
		1	Yes	
C1_2	Trastorno maníaco-depresivo, bipolar o manía	0	No	
		1	Yes	
C1_3	Crisis o ataques de pánico	0	No	
		1	Yes	
C1_4	Problemas de ansiedad (nervios, preocupaciones, temores, obsesiones)	0	No	
		1	Yes	
C1_5	Problemas con el consumo de alcohol, abuso de alcohol o dependencia del alcohol	0	No	
		1	Yes	
C1_6	Problemas con consumo de drogas como el consumo de drogas ilegales y el consumo de medicamentos psicotrópicos (tranquilizantes, estimulantes, pastillas para el dolor) sin receta	0	No	
		1	Yes	
C1_7	Cualquier otro problema psicológico serio (describa a continuación)	0	No	
		1	Yes	
C1_8	Ningún problema psicológico	0	No	
		1	Ningún problema	
C3	**En los últimos 14 días, ¿con qué frecuencia le han supuesto una molestia los siguientes problemas?**			
C3_1	Sentirse nervioso/a, angustiado/a o muy tenso/a	1	Nunca	
		2	Varios días	
		3	Más de la mitad de los días	
		4	Casi cada día	
C3_2	Ser incapaz de dejar de preocuparse o de controlar la preocupación	1	Nunca	
		2	Varios días	
		3	Más de la mitad de los días	
		4	Casi cada día	
C3_3	Preocuparse demasiado por diferentes cuestiones	1	Nunca	
		2	Varios días	
		3	Más de la mitad de los días	
		4	Casi cada día	
C3_4	Tener problemas para relajarse	1	Nunca	
		2	Varios días	
		3	Más de la mitad de los días	
		4	Casi cada día	
C3_5	Estar tan inquieto/a que le resulta difícil permanecer sentado/a	1	Nunca	
		2	Varios días	
		3	Más de la mitad de los días	
		4	Casi cada día	

		4	Casi cada día
C3_6	Enfadarse o irritarse con facilidad	1	Nunca
		2	Varios días
		3	Más de la mitad de los días
		4	Casi cada día
C3_7	Sentir miedo de que algo terrible pueda ocurrir	1	Nunca
		2	Varios días
		3	Más de la mitad de los días
		4	Casi cada día
C4	**En los últimos 30 días, ¿cuánta molestia ha sentido por:**		
C4_1	sentir o actuar repentinamente como si la experiencia estresante estuviera sucediendo nuevamente (como si estuviera reviviéndola)?	1	Nada
		2	Un poco
		3	Moderadamente
		4	Bastante
		5	Extremadamente
C4_2	evitar claves o recordatorios externos de la experiencia estresante (por ejemplo, personas, lugares, conversaciones, actividades, objetos o situaciones)?	1	Nada
		2	Un poco
		3	Moderadamente
		4	Bastante
		5	Extremadamente
C4_3	sentirse distante o enajenado/a de otras personas?	1	Nada
		2	Un poco
		3	Moderadamente
		4	Bastante
		5	Extremadamente
C4_4	irritabilidad, explosiones de rabia o actuar agresivamente?	1	Nada
		2	Un poco
		3	Moderadamente
		4	Bastante
		5	Extremadamente
C5	¿aproximadamente cuántos ataques de pánico tuvo en los últimos 30 días?	.	(0-200)
C10	En los últimos 30 días ¿alguna vez ha deseado estar muerto/a o poder dormirse y no volver a despertar?	0	No
C10		1	Yes
C11	En los últimos 30 días ¿alguna vez ha deseado estar muerto/a o poder dormirse y no volver a despertar?	0	No
C11		1	Yes
C12	Aproximadamente cuántos de los últimos 30 días (deseos o pensamientos)		(0-30)
C13	En los últimos 30 días ¿alguna vez ha pensado en cómo podría suicidarse (por ejemplo, tomando pastillas o tirándose por una ventana) o ha planeado hacerlo?	0	No
C13		1	Yes
C14	¿cuántos de los últimos 30 días ha pensado en cómo podría suicidarse?	0	0-30
C15	En los últimos 30 días, ¿ha cometido un intento de suicidio (es decir, se ha autolesionado deliberadamente con al menos alguna intención de morir)? (sí/no)	1	Yes
		0	No
D1	**¿Alguna vez en la vida ha recibido ayuda psicológica o medicación por un problema emocional o relacionado con el consumo de sustancias?**		
D1_1	Ayuda Psicológica	1	Yes
		0	No
D1_2	Medicación	1	Yes
		0	No
D2	**¿Alguna vez en los últimos 12 meses ha recibido ayuda psicológica o medicación por un problema emocional o relacionado con el consumo de sustancias?**		
D2_1	Ayuda Psicológica	0	No
		1	Yes
D2_2	Medicación	0	No
		1	Yes
D5	Desde de la llegada de la pandemia del coronavirus a España, ¿Ha habido algún momento en el que sintió que podría necesitar ayuda psicológica o medicación adicional para algún problema psicológico o relacionado con el consumo de sustancias?	0	No
D5		1	Yes
D6	Desde de la llegada de la pandemia del coronavirus a España, ¿Ha habido algún momento en el que sintió que podría necesitar ayuda psicológica o medicación para algún problema psicológico o relacionado con el consumo de sustancias? (sí/no)	0	No
D6		1	Yes

1.3 Scales used to measure participants

VARIABLES CREADAS A PARTIR DE LOS ÍTEMS DEL CUESTIONARIO			
Infección por covid:			
test_or_diag	Prueba positiva (B2=1) or diagnóstico médico COVID (B3=1)	0	No
		1	Yes
hosp	¿Hospitalización por COVID?	0	No
		1	Yes
inf_status	Infection status	1	having been hospitalized for COVID
		2	positive COVID-19 test or medical Diagnostic
		3	-none of the above
Trastorno mental previo:			
mental_pre_mood	lifetime mood disorder	0	no lifetime mood disorder
		1	lifetime mood disorder
mental_pre_anx	lifetime anxiety disorder	0	no lifetime anxiety disorder
		1	lifetime anxiety disorder
mental_pre_sub	lifetime substance use disorder	0	no lifetime substance use diso
		1	lifetime substance use disorde
mental_pre_oth	other lifetime mental disorder	0	no other lifetime mental disor
		1	other lifetime mental disorder
any_mental_pre	Any previous mental problem?	1	Yes
		0	No
Trastorno mental actual			
score_gad	Puntuación GAD-7 contínua (rango 0-21)	.	
gad_2w	Current GAD (Score_gad >=10)	1	Yes
gad_2w		0	No
panic_30dd	Current Panic Attacks (>=1 panic attack)	1	Yes
		0	No
score_ptsd		.	
ptsd_30dd	Current PTSD (score_ptsd >=7)	1	Yes
		0	No
stb_30dd	Suicide Thoughts and Behaviours Gradient	0	None
		1	passive ideation
		2	active ideation
		3	suicide plan
		4	suicide attempt
Treatment Use			
TX_LT	Mental Health service use lifetime	0	No
		1	Yes
TX_12m	Mental Health service use last 12 months	0	No
		1	Yes
treat_need	Perceived need for treatment	0	No
		1	Yes

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